

**Supplementary Material for: Low-temperature Stress Alters Gene Expression of the
Kidney Transcriptome in *Takifugu rubripes***

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Supplementary Note 1: The GitHub repository for this project is available at https://github.com/Amisor/KidneyTranscriptome_TakifuguRubripes. It includes instructions for navigating the directories and files, as well as documented R code and datasets.

Supplementary Table 1. Total number of collected reads and mapped reads per group.

Reads (full collection)			Mapped reads		
Per Group	Reads	% of Total	Per Group	Reads	% of Total
Control	67,958,297	49.6	Control	28,115,895	49.4
Low	69,051,768	50.4	Low	28,845,172	50.6
Total	137,010,065	100	Total	56,961,067	100

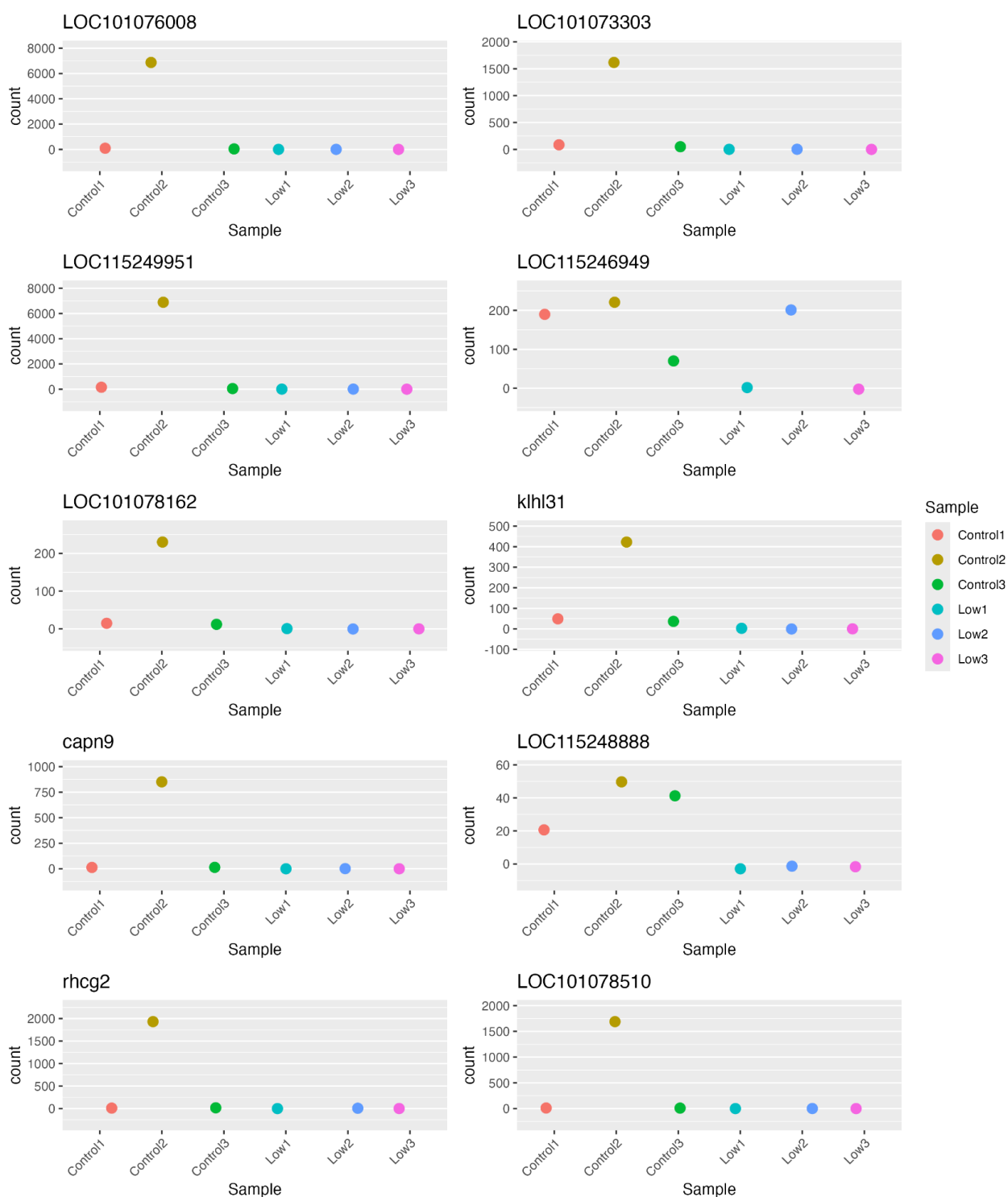
Supplementary Table 2. Number of mapped reads and their percentage relative to the total reads per sample and per group.

Category	Sample/Group	Reads	Mapped Reads	% of Mapped Reads
Per Sample	Control 1	23,255,676	9,789,875	42.1
	Control 2	21,908,548	9,080,028	41.4
	Control 3	22,794,073	9,245,992	40.6
	Low 1	23,240,071	9,725,140	41.8
	Low 2	22,253,023	9,363,929	42.1
	Low 3	23,558,674	9,756,103	41.4
Per Group	Control	67,958,297	28,115,895	41.4
	Low	69,051,768	28,845,172	41.8
Total	Total	137,010,065	56,961,067	41.6

Supplementary Table 3. Number of genes with NA values in different columns of the DESeq2 results. It was confirmed that all genes with a baseMean of zero correspond to those with NA values in both the p-value and log2FoldChange columns. For further details, refer to the script *01_DESeq.R* available at the [GitHub repository](#) of the project.

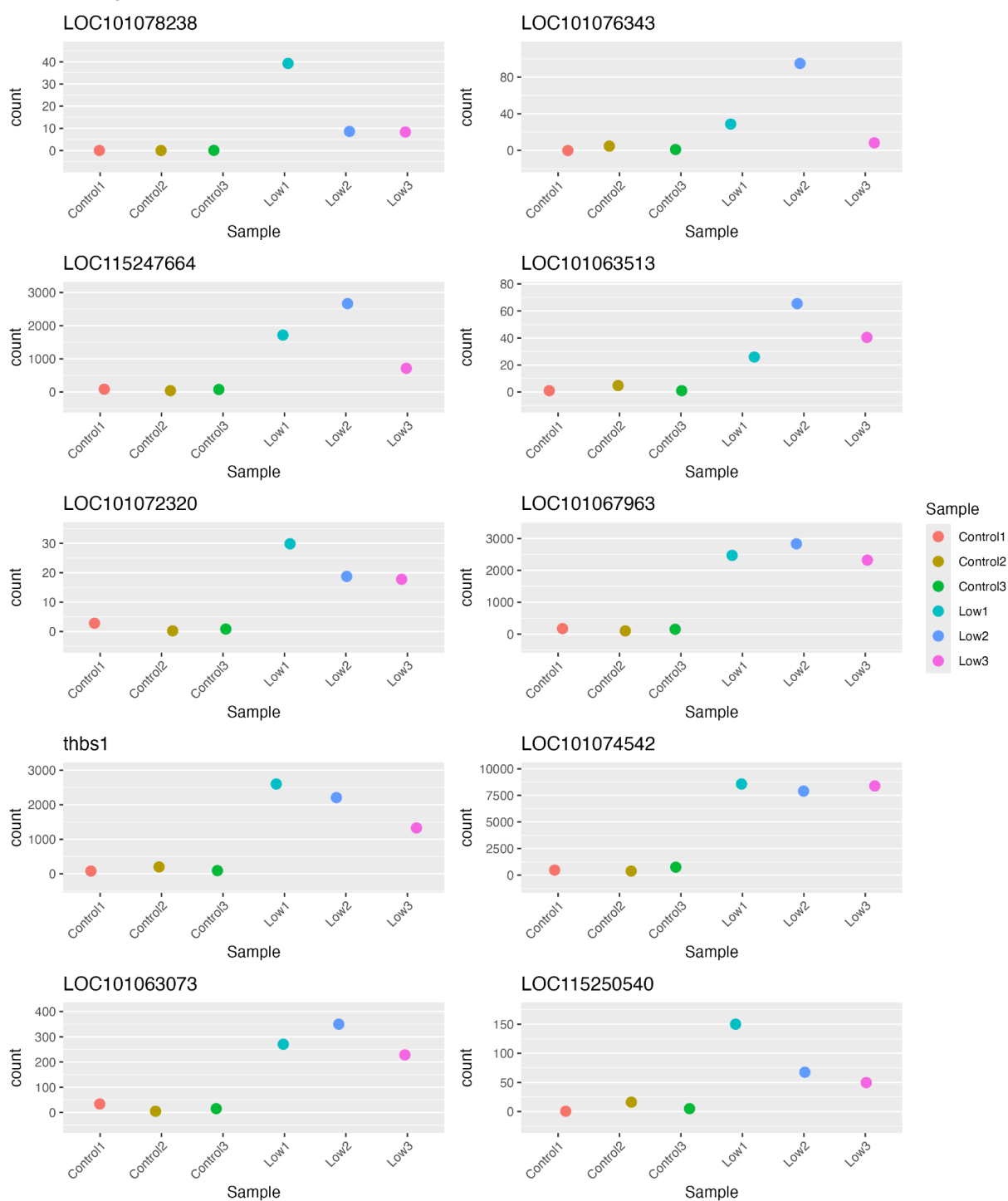
	Number
All genes	26,923
Genes with zero in baseMean	4,044
Genes with NA in log2FoldChange	4,044
Genes with NA in p-value	4,044
Genes with NA in padj	9,757
Genes with NA in log2FoldChange and p-value	4,044
Genes with NA in log2FoldChange and padj	4,044
Genes with NA in p-value and padj	4,044
Genes with NA in log2FoldChange, p-value and padj	4,044

Top 10 Downregulated Genes: Normalized Counts per Sample

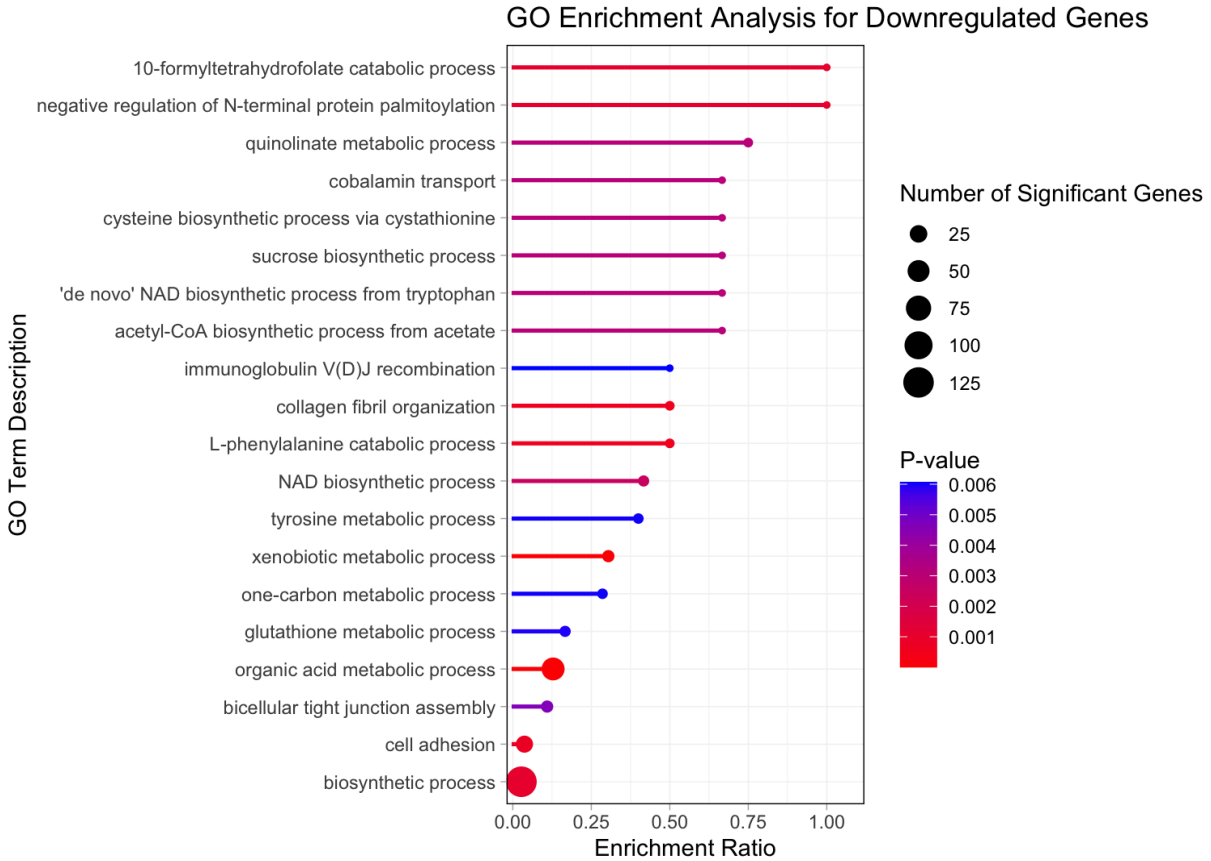


Supplementary Figure 1. Normalized count for the top 10 downregulated genes in different samples.

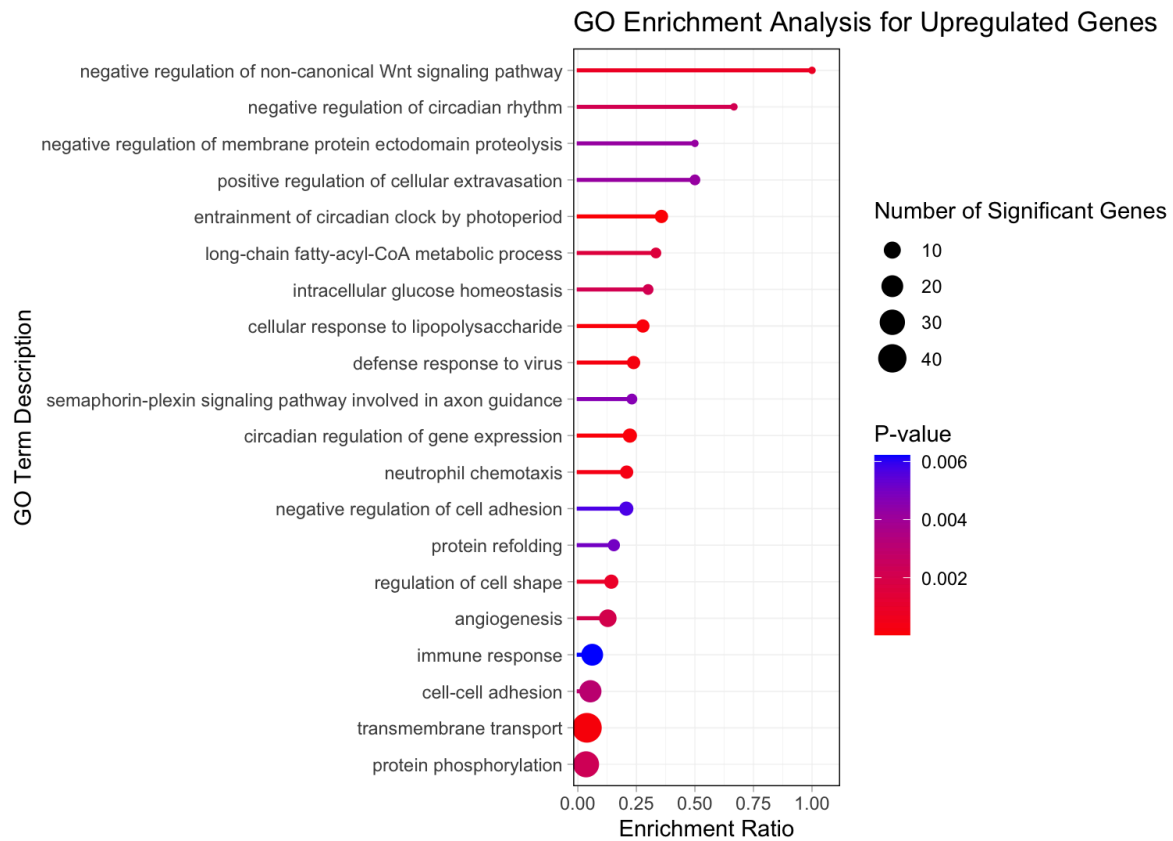
Top 10 Upregulated Genes: Normalized Counts per Sample



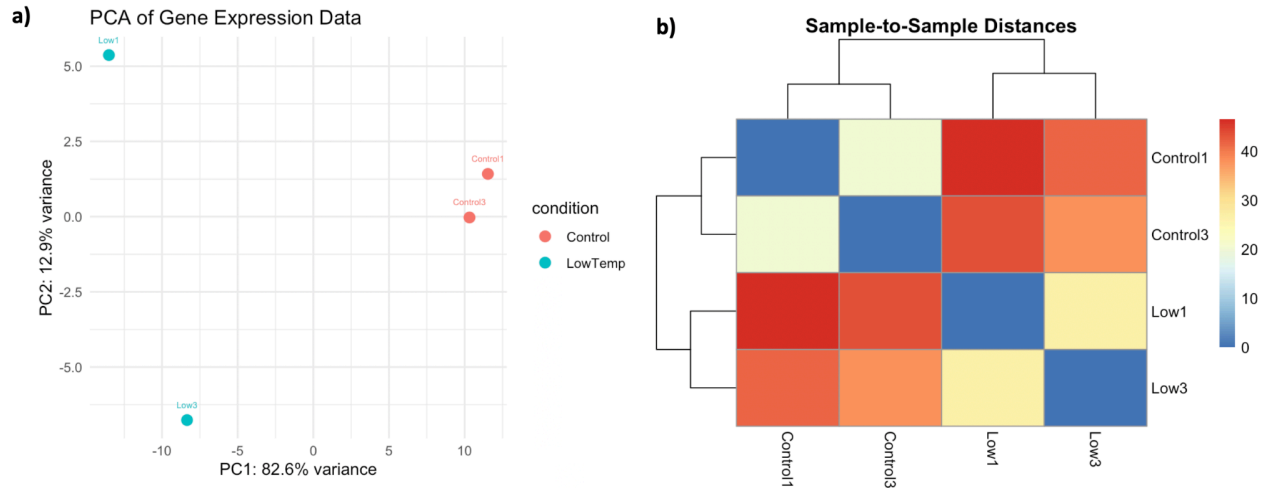
Supplementary Figure 2. Normalized count for the top 10 upregulated genes in different samples.



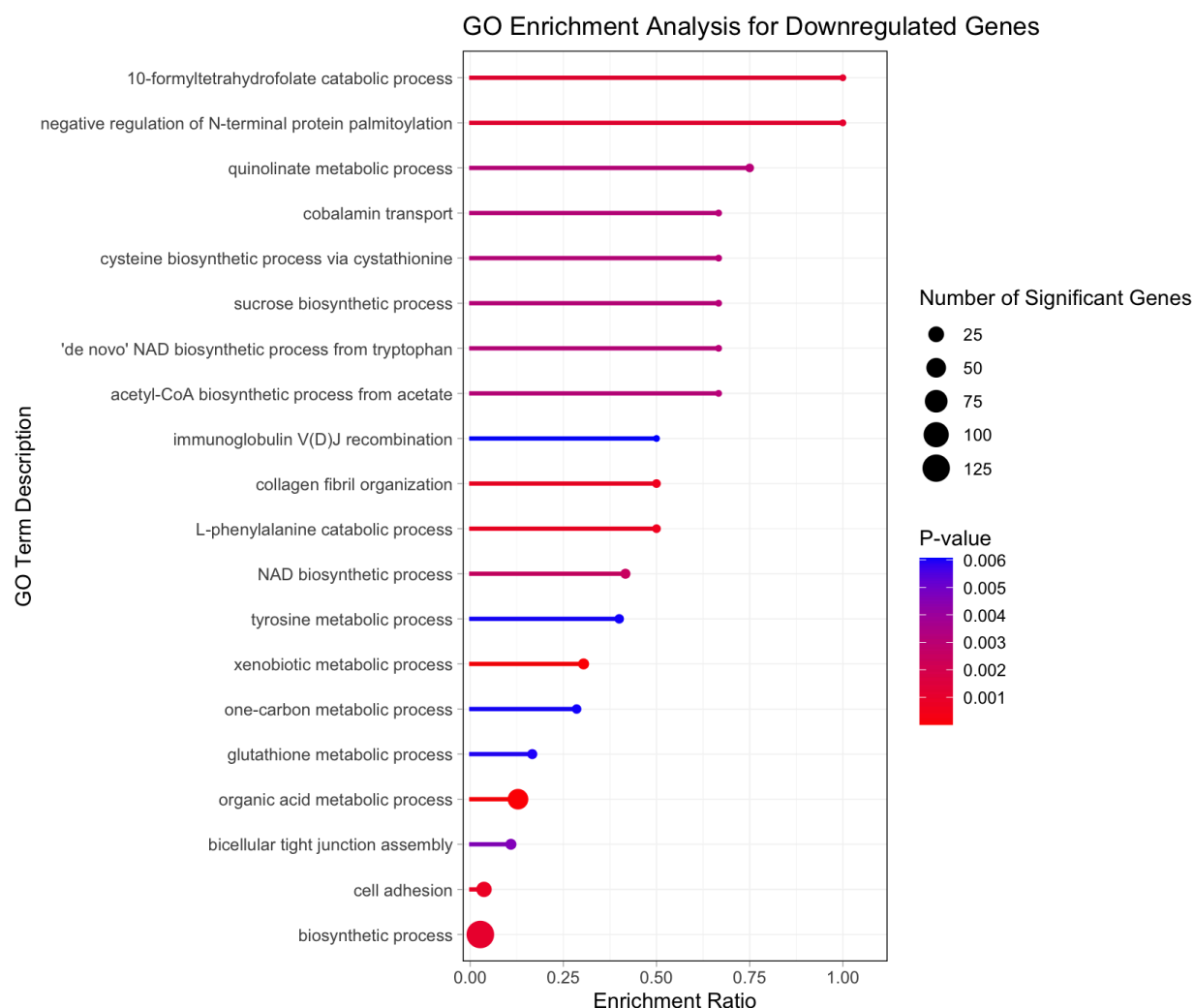
Supplementary Figure 3. Go enrichment analysis for downregulated genes. The size of each dot corresponds to the number of significant genes associated with the term, while the color represents the p-value, with more statistically significant terms highlighted in red.



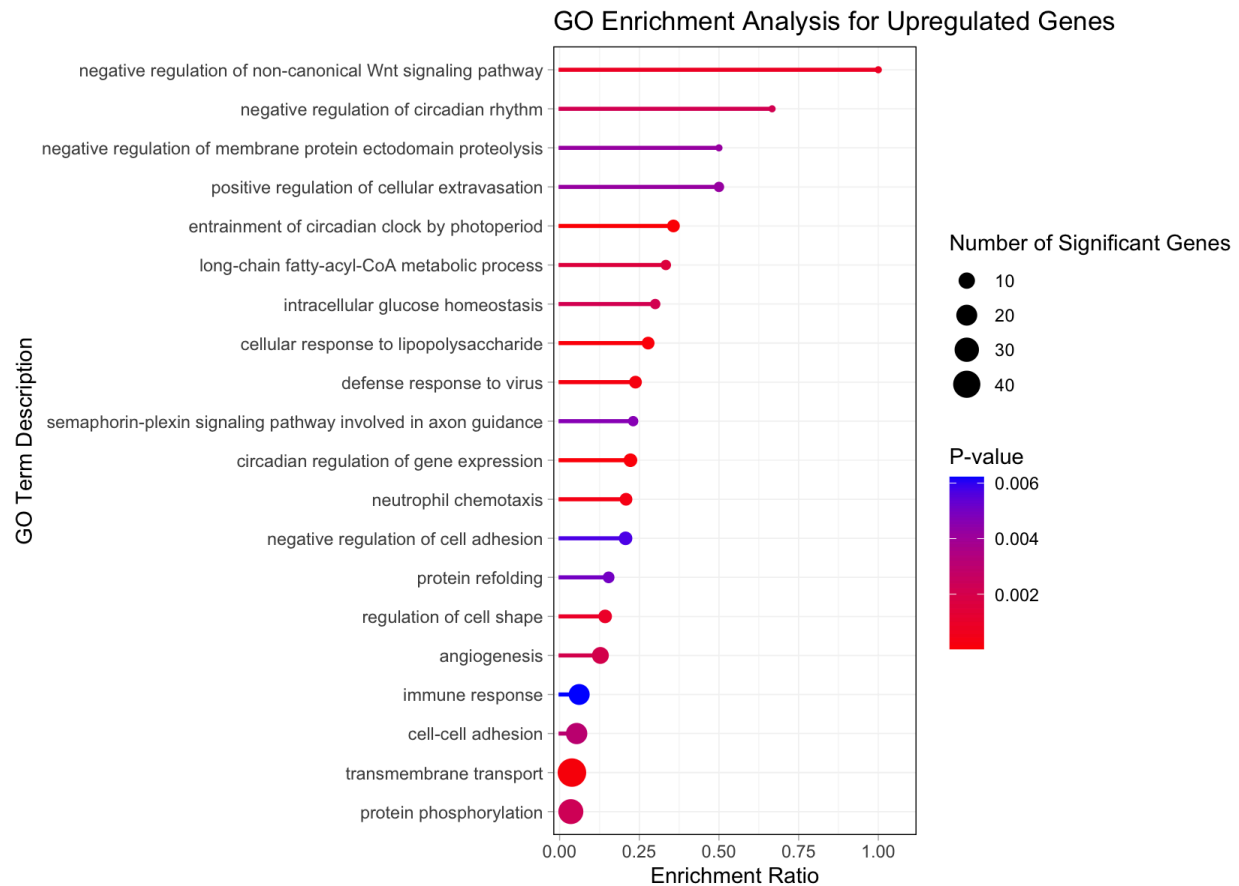
Supplementary Figure 4. Go enrichment analysis for upregulated genes. The size of each dot corresponds to the number of significant genes associated with the term, while the color represents the p-value, with more statistically significant terms highlighted in red.



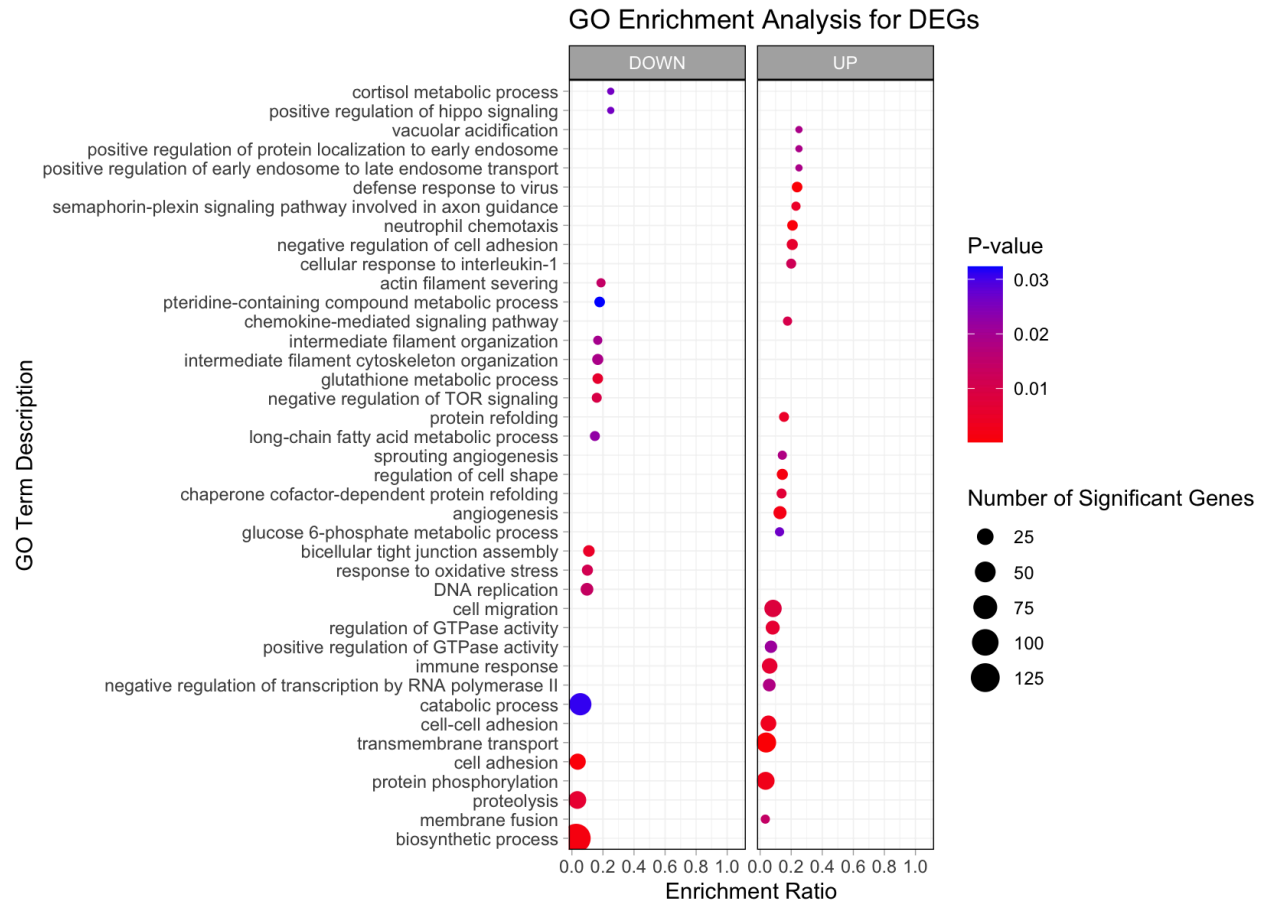
Supplementary Figure 5. a) Principal Component Analysis (PCA) plot of gene expression data without *Control2* and *Low2* samples. Samples are colored by group. **b)** Heatmap of sample-to-sample distances. Euclidean distances between normalized gene expression profiles (rlog-transformed) were taken. Clustering of the rows and columns reveals similarities between samples.



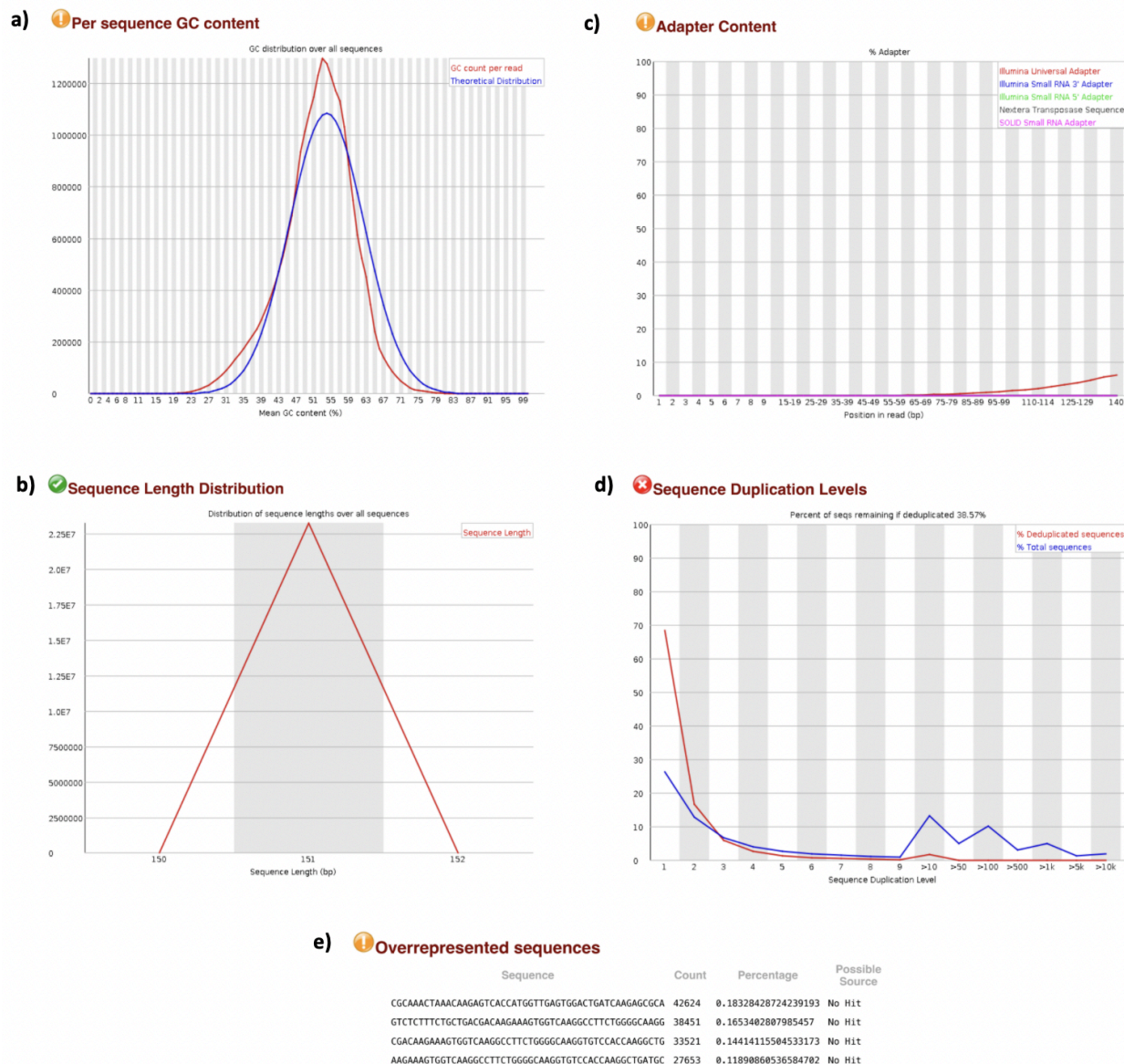
Supplementary Figure 6. Go enrichment analysis for downregulated genes without *Control2* and *Low2* samples. The size of each dot corresponds to the number of significant genes associated with the term, while the color represents the p-value, with more statistically significant terms highlighted in red.



Supplementary Figure 7. Go enrichment analysis for upregulated genes without *Control2* and *Low2* samples. The size of each dot corresponds to the number of significant genes associated with the term, while the color represents the p-value, with more statistically significant terms highlighted in red.



Supplementary Figure 8. Go enrichment analysis for upregulated and downregulated genes without *Control2* and *Low2* samples. The size of each dot corresponds to the number of significant genes associated with the term, while the color represents the p-value, with more statistically significant terms highlighted in red.



Supplementary Figure 9. FastQC analysis of *Controll* sample. **a)** Per sequence GC content plot, which raises a warning due to a higher than expected GC content. **b)** Sequence Length Distribution plot, showing that all reads are 151 bp. **c)** Adapter Content plot, identifying the presence of Illumina Universal Adapter at the last positions of the sequences. **d)** Sequence Duplication Levels plot, which fails due to highly abundant and lowly abundant transcripts. Refer to the [GitHub repository](#) for the complete FastQC analysis of all samples from the Control and Low-Temperature groups.